ANIKEF

DESCRIPTION:
ANIKEF STERILE 750MG: A cream coloured powder packed in a 10mL amber glass vial
ANIKEF STERILE 1.5G: A cream coloured powder packed in a 20mL amber glass vial

COMPOSITION:
ANIKEF STERILE 750MG: Each vial contains Cefuroxime Sodium equivalent to Cefuroxime 750 mg
ANIKEF STERILE 1.5G: Each vial contains Cefuroxime Sodium equivalent to Cefuroxime 1.5 g

PHARMACODYNAMICS: Cefuroxime is bactericidal and has a similar spectrum of antimicrobial action and pattern of resistance to cephamandole. It is more resistant to hydrolysis by beta-lactamases than cephamandole, and is therefore more active against some beta-lactamase-producing strains. The bactericidal action of Cefuroxime results from inhibition of cell wall synthesis by binding to essential target proteins.

PHARMACOKINETICS: Cefuroxime is poorly absorbed from the gastrointestinal tract and is given by intramuscular or intravenous injection as the sodium salt. Peak plasma concentrations of 26 to 34µg per ml have been achieved about 45 minutes after an intramuscular dose of 750 mg with measurable amounts present 8 hours after a dose; after a dose of 1 g peak concentrations are 32 to 40 µg per ml. About 33% of cefuroxime in the circulation is bound to plasma proteins. The plasma half-life is about 70 minutes and is prolonged in patients with renal impairment and in neonates. Cefuroxime is widely distributed in the body including pleural fluid, sputum, bone, synovial fluid, and aqueous humour. It diffuses across the placenta and has been detected in breast milk, but only achieves therapeutic concentrations in the CSF when the meninges are inflamed. Most of a dose of cefuroxime is excreted unchanged, by glomerular filtration and renal tubular secretion, within 24 hours, and the majority within 6 hours; high concentrations are achieved in the urine. Probenecid competes for renal tubular secretion with cefuroxime resulting in higher and more prolonged plasma concentrations of cefuroxime. Small amounts of cefuroxime are excreted in bile.

INDICATIONS: Respiratory tract infections, eg. acute and chronic bronchitis, infected bronchiectasis, bacterial pneumonia, lung abscess and postoperative chest infections.
Ear, nose and throat infections, eg, sinusitis, tonsillitis and pharyngitis.
Urinary tract infections, eg. acute and chronic pyelonephritis, cystitis and asymptomatic bacteriuria.
Soft-tissue infections, eg. cellulitis, erysipelas, peritonitis and wound infections.
Bone and joint infections, eg. osteomyelitis and septic arthritis.
Obstetric and gynaecological infections, pelvic inflammatory diseases.
Gonorrhoea particularly when penicillin is unsuitable.
Other infections including septicaemia and meningitis.
Prophylaxis against infection in abdominal, pelvic, orthopaedic, cardiac, pulmonary, oesophageal and vascular surgery where there is increased risk from infection.

RECOMMENDED DOSAGE: In the treatment of susceptible infections. It is administered by deep intramuscular injection, by slow intravenous injection over 3 to 5 minutes, or by intravenous infusion. A usual dose is the equivalent of 750 mg of cefuroxime every 8 hours but in more severe infections up to 1.5 g may be given 6 hourly. Infants and children can be given 30 to 60 mg per kg body-weight daily, increased to 100 mg per kg daily if necessary, given in 3 or 4 divided doses. Neonates may be given similar total daily doses but administered in 2 or 3 divided doses.
For the treatment of meningitis due to sensitive strains of bacteria, cefuroxime sodium may be administered intravenously in doses equivalent to cefuroxime 3 g every 8 hours. Infants and children are given 200 to 240 mg per kg daily in 3 or 4 divided doses, which may be decreased to 100 mg per kg daily after 3 days. For neonates, a dose of 100 mg per kg daily, decreased to 50 mg per kg daily when indicated, has been recommended.
In the treatment of penicillin-resistant gonorrhoea a single dose of 1.5 g by intramuscular injection has been suggested. The dose may be divided between 2 injection sites, and may be given in conjunction with probenecid by mouth. For surgical infection prophylaxis, usually in a dose equivalent to 1.5 g of cefuroxime intravenously prior to the procedure followed by 750 mg intravenously or intramuscularly every 8 hours for up to 24 to 48 hours. For total joint replacement 1.5 g of cefuroxime powder may be mixed with the methylmethacrylate cement.

Administration: IM: Add 1 ml water for injections to 250 mg Cefuroxime Sodium or 3 ml water for injections to 750 mg Cefuroxime Sodium. Shake gently to produce an opaque suspension.
IV: Dissolve Cefuroxime Sodium in water for injections using at least 2 ml for 250 mg, at least 6 ml for 750 mg or 15 ml for 1.5 g. For short IV infusion (e.g. up to 30 min.), 1.5 g may be dissolved in 50 ml water for injections. These solutions may be given directly into the vein or introduced into the tubing of the giving set if the patient is receiving parenteral fluids.
Cefuroxime Sodium is compatible with the more commonly use IV fluids.
Solutions range from light yellow to amber depending on concentration, diluent and storage condition used. Within the stated recommendations, product potency is not adversely affected by such colour variations.

ROUTE OF ADMINISTRATION: For i.m and i.v injection

CONTRAINDICATIONS: Hypersensitivity to cephalosporin antibiotics. Not to be used in patients with known hypersensitivity to penicillin.

WARNING & PRECAUTIONS: About 10% of penicillin-sensitive patients will also be allergic to cephalosporins and therefore great care should be taken if cefuroxime is to be given to patients who are known to be allergic to penicillins. Care is also necessary in patients with known histories of allergy. Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with cefuroxime, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, cefuroxime must be discontinued immediately and appropriate alternative therapy should be instituted.
Cefuroxime Sodium should be given with caution to patients with renal impairment: a dosage reduction may be necessary. The concomitant use of diuretics such as frusemide and nephrotoxic antibiotics such as gentamicin may increase the risk of kidney damage. Renal and haematological status should be monitored especially during prolonged and high-dose therapy.

INTERACTION WITH OTHER MEDICAMENTS: Probenecid – Probenecid decreases renal tubular secretion of those cephalosporins excreted by this mechanism, resulting in increased and prolonged cephalosporin, serum concentrations, prolonged elimination half-life, and increased risk of toxicity. Besides cefoperazone, ceftadizime, or ceftriaxone, other cephalosporins and probenecid might be used concurrently in the treatment of infections such as sexually transmitted disease (STDs) or other infections, in which high and/or prolonged antibiotic serum and tissue concentrations are required.

INCOMPATIBILITIES: The admixture of beta-lactam antibacterial (penicillins and cephalosporins) and amino-glycosides may result in substantial mutual inactivation. If they are administered concurrently, they should be administered in separate sites. Do not mix them in the same intravenous bag or bottle.
Caution: Cefuroxime sodium may be incompatible with aminoglycosides. Darkening of the solution may occur on storage.

USE IN PREGNANCY & LACTATION:
Use in Pregnancy*:* At doses 60 times the human dose, rabbits and mice did not reveal any evidence of impaired fertility or harm to the foetus. Since there are no adequate well controlled studies on pregnant women, this drug should be used in pregnancy only if clearly needed.
Use in Lactation*:* If cefuroxime is given to a nursing mother, caution should be exercised since cefuroxime is excreted in human milk.

SIDE EFFECTS: Allergic reactions, including skin rashes, urticaria, eosinophilia, fever, reactions resembling serum sickness and anaphylaxis may occur in patients receiving cefuroxime sodium, especially if they are hypersensitive to penicillins or other cephalosporins. Neutropenia, leucopenia, thrombocytopenia, and haemolytic anaemia have occasionally been reported. Nephrotoxicity has occurred particularly in patients with existing renal impairment or in patients receiving concomitant nephrotoxic drugs.
Transient increases in liver enzyme values, including serum aspartate and alkaline phosphatase, have been reported. Neurological disturbances including encephalopathy have occurred occasionally. There may be pain at the injection site following intramuscular administration and thrombophlebitis has occurred following intravenous infusion, usually of more than 6 g daily for more than 3 days. Gastro-intestinal adverse effects have been reported rarely. Prolonged use may result in overgrowth of non-susceptible organisms and, as with other broad-spectrum antibiotics, pseudomembranous colitis may develop.

SYMPTOMS AND TREATMENT OF OVERDOSE: Overdosage: Serum levels of cefuroxime are reduced by dialysis. Overdosage of cephalosporins can cause cerebral irritation leading to convulsions.

STORAGE CONDITIONS: Store below 30°C. Protect from light.
If not used immediately after preparation, the constituted solution should be stored below 25°C and used within 5 hours and if refrigerated, it should be used within 48 hours. KEEP OUT OF REACH OF CHILDREN. *JAUHKAN DARIPADA KANAK-KANAK*.

SHELF LIFE: Please refer to outer package

PACK SIZE: Pack in a box of 1 and 25 vials

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59,Kawasan 3, Taman Klang Jaya,
41200 Klang, Selangor, Malaysia.